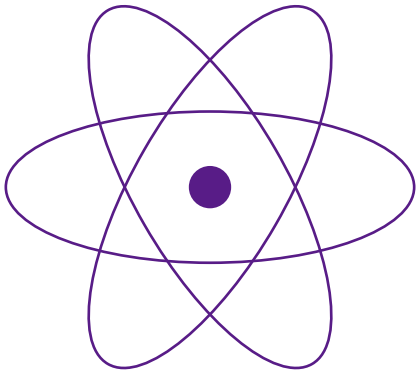
SPhOH

Saudi Physics Olympiad Hub



Trigonometry I

Detailed Solutions Manual

Olympiad Training ▪ Math

SPhOH Team

First Edition ▪ 2026

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Chapter 1

Trigonometry Solutions

1.1 Trigonometric Identities

Solution to Problem 1

The Fundamental Identity

From the definition of the unit circle or using a right triangle with hypotenuse $r = 1$, let the side opposite to angle x be y and the adjacent side be x (i.e., $\sin x = y$ and $\cos x = x$). The Pythagorean theorem directly yields:

$$x^2 + y^2 = 1 \implies \sin^2 x + \cos^2 x = 1$$

This is the fundamental trigonometric identity.

Solution to Problem 2

A Tangent Identity

We start with the definition of the tangent function:

$$\tan x = \frac{\sin x}{\cos x}$$

Squaring both sides:

$$\tan^2 x = \frac{\sin^2 x}{\cos^2 x}$$

Adding 1 to both sides and substituting $\sin^2 x = 1 - \cos^2 x$ from the fundamental identity:

$$1 + \tan^2 x = 1 + \frac{\sin^2 x}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

Since $\sin^2 x + \cos^2 x = 1$, we obtain the required result:

$$1 + \tan^2 x = \frac{1}{\cos^2 x}$$

Solution to Problem 3

Complementary Angles

(a) The sum of the two acute angles in a right-angled triangle is $90°$:

$$A + B + 90° = 180° \implies A + B = 90°$$

(b) From the right triangle, the sine of angle A is the opposite over the hypotenuse, while the cosine of angle B shares the adjacent side to B (which is opposite to A) and the same hypotenuse:

$$\sin A = \frac{\text{opp of } A}{\text{hyp}} = \frac{\text{adj of } B}{\text{hyp}} = \cos B$$

(c) Since $A = 90° - B$, substituting this into the previous result gives:

$$\cos B = \sin(90° - B)$$

(d) Geometrically, interchanging the roles of the two acute angles in a right triangle means that the side opposite one is adjacent to the other, leading to the equality of sine and cosine ratios for complementary angles.

1.2 Right-Triangle Trigonometry

Solution to Problem 4 — The 3–4–5 Triangle

The hypotenuse is the longest side $c = 5$. Choosing the acute angle θ opposite to side 3:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{3}{5} = 0.6, \qquad \csc\theta = \frac{5}{3}$$
$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{4}{5} = 0.8, \qquad \sec\theta = \frac{5}{4}$$
$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{4} = 0.75, \qquad \cot\theta = \frac{4}{3}$$

Solution to Problem 5 — A Right Triangle with Known Sides

(a) Using the Pythagorean theorem (hypotenuse $c = 16$ and $AC = 12$):

$$BC = \sqrt{16^2 - 12^2} = \sqrt{256 - 144} = \sqrt{112} = 4\sqrt{7}\ \text{cm} \approx 10.58\ \text{cm}$$

(b) $\sin B = \frac{\text{opp of } B}{\text{hyp}} = \frac{AC}{AB} = \frac{12}{16} = \frac{3}{4} = 0.75.$

(c) $\cos B = \frac{\text{adj of } B}{\text{hyp}} = \frac{BC}{AB} = \frac{4\sqrt{7}}{16} = \frac{\sqrt{7}}{4} \approx 0.661.$

(d) Angle C is the right angle itself ($C = 90°$). Therefore:

$$\tan C = \tan 90° = \frac{\sin 90°}{\cos 90°} = \frac{1}{0} \implies \text{undefined (tends to infinity)}$$

(e) As an angle approaches $90°$, its cosine tends to zero while its sine tends to one, making the tangent ratio undefined.

Solution to Problem 6

Working Backwards from Sine

(a) The hypotenuse is $XY = 10$ cm. Since the side opposite to angle X is YZ, and we are given $\sin X = \frac{YZ}{XY} = \frac{3}{5}$:

$$YZ = 10 \times \frac{3}{5} = 6 \text{ cm}$$

(b) To find the adjacent side XZ, we use the Pythagorean theorem:

$$XZ = \sqrt{XY^2 - YZ^2} = \sqrt{10^2 - 6^2} = \sqrt{64} = 8 \text{ cm}$$

(c) $\cos X = \frac{XZ}{XY} = \frac{8}{10} = 0.8$.

(d) For angle Y: the opposite side is $XZ = 8$ and the adjacent side is $YZ = 6$:

$$\tan Y = \frac{\text{opp of } Y}{\text{adj of } Y} = \frac{XZ}{YZ} = \frac{8}{6} = \frac{4}{3}$$

Solution to Problem 7

The 30–60–90 Triangle

(a) The side opposite the $30°$ angle is half the hypotenuse:

$$\text{Shorter side} = \frac{1}{2} \times 12 = 6 \text{ cm}$$

The side adjacent to the $30°$ angle (opposite the $60°$ angle) is $\sqrt{3}$ times the shorter side:

$$\text{Longer side} = 6\sqrt{3} \text{ cm} \approx 10.39 \text{ cm}$$

(b) Area of the triangle:

$$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 6 \times 6\sqrt{3} = 18\sqrt{3} \text{ cm}^2 \approx 31.18 \text{ cm}^2$$

(c) Because any two triangles of this type are similar (equal angles), their side ratios remain constant regardless of size.

Solution to Problem 8

Exact Trigonometric Values

(a) $\sin 30^\circ = \frac{1}{2}$

(b) $\cos 60^\circ = \frac{1}{2}$

(c) $\sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

(d) $\cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

(e) $\tan 45^\circ = 1$

Solution to Problem 9

The Right Isosceles Triangle

(a) Using the Pythagorean theorem on the two equal legs a:

$$c = \sqrt{a^2 + a^2} = \sqrt{2a^2} = a\sqrt{2}$$

(b) Since the two legs are equal, the two acute angles are equal and sum to 90°, meaning both are 45°.

(c)

$$\sin 45^\circ = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \cos 45^\circ = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \tan 45^\circ = \frac{a}{a} = 1$$

1.3 Special Triangles and Geometry

Solution to Problem 10

Area of an Equilateral Triangle

Dropping an altitude from the vertex of the triangle to its base a bisects the base into two segments of length $\frac{a}{2}$, forming two right triangles. The height h is found via the Pythagorean theorem:

$$h = \sqrt{a^2 - \left(\frac{a}{2}\right)^2} = \sqrt{a^2 - \frac{a^2}{4}} = \sqrt{\frac{3a^2}{4}} = \frac{\sqrt{3}}{2}a$$

Total area of the triangle:

$$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times a \times \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a^2$$

Solution to Problem 11 — Dividing an Equilateral Triangle

Side length $a = 10$ cm.

(a) Height:

$$h = \frac{\sqrt{3}}{2} \times 10 = 5\sqrt{3} \text{ cm} \approx 8.66 \text{ cm}$$

(b) Area:

$$A = \frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3} \text{ cm}^2 \approx 43.30 \text{ cm}^2$$

(c) In the resulting right triangle: the side opposite the 30° angle is 5, the side opposite the 60° angle is $5\sqrt{3}$, and the hypotenuse is 10:

$$\sin 30^\circ = \cos 60^\circ = \frac{5}{10} = \frac{1}{2}, \quad \cos 30^\circ = \sin 60^\circ = \frac{5\sqrt{3}}{10} = \frac{\sqrt{3}}{2}$$

1.4 Similarity and Trigonometric Ratios

Solution to Problem 12 — Similar Right Triangles

(a) Using the Pythagorean theorem in triangle ABC:

$$BC = \sqrt{6^2 - 4^2} = \sqrt{36 - 16} = \sqrt{20} = 2\sqrt{5} \text{ m} \approx 4.47 \text{ m}$$

(b) $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AC}{AB} = \frac{4}{6} = \frac{2}{3}$.

(c) Due to triangle similarity and equal angle $A = X$:

$$\cos X = \cos A = \frac{2}{3}$$

(d) Because trigonometric ratios depend entirely on side ratios rather than absolute lengths; scaling a triangle scales all sides by a common factor that cancels out in numerator and denominator.

Solution to Problem 13 — Same Angle, Different Size

(a) The triangles are similar, so their tangents are equal:

$$\tan\theta = \frac{\text{opposite}_1}{\text{adjacent}_1} = \frac{3}{4} = 0.75$$

For the second triangle:

$$\frac{\text{opposite}_2}{\text{adjacent}_2} = 0.75 \implies \text{opposite}_2 = 0.75 \times 10 = 7.5 \text{ cm}$$

(b) Because a constant angle guarantees a constant ratio between the two sides (tangent), allowing the unknown to be found directly via proportion without calculating θ.

1.5 Trigonometry in Physics

Solution to Problem 14

Resolving a Force

Force $F = 10$ N at an angle $\theta = 30°$:

(a) Horizontal component:

$$F_x = F\cos 30° = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3} \text{ N} \approx 8.66 \text{ N}$$

(b) Vertical component:

$$F_y = F\sin 30° = 10 \times \frac{1}{2} = 5 \text{ N}$$

(c) Reconstructing the force using the Pythagorean theorem:

$$F_{\text{net}} = \sqrt{F_x^2 + F_y^2} = \sqrt{(5\sqrt{3})^2 + 5^2} = \sqrt{75 + 25} = \sqrt{100} = 10 \text{ N}$$

with direction $\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = \tan^{-1}\left(\frac{5}{5\sqrt{3}}\right) = 30°$, fully reconstructing the original force.

Solution to Problem 15

Velocity Components

Velocity $v = 20$ m/s at an angle $\theta = 60°$:

(a) Horizontal component:

$$v_x = v\cos 60° = 20 \times \frac{1}{2} = 10 \text{ m/s}$$

(b) Vertical component:

$$v_y = v\sin 60° = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \text{ m/s} \approx 17.32 \text{ m/s}$$

(c) The vertical component is larger because the angle is measured relative to the horizontal, making the cosine of $60°$ smaller than its sine ($0.5 < 0.866$).

Solution to Problem 16

Recovering an Unknown Direction

Force $F = 50$ N, with horizontal component $F_x = 30$ N.

(a) Related equation:

$$F_x = F\cos\theta \implies 30 = 50\cos\theta \implies \cos\theta = \frac{30}{50} = 0.6$$

(b) Finding the angle:

$$\theta = \cos^{-1}(0.6) \approx 53.13^\circ$$

(c) Finding the vertical component:

$$F_y = F\sin\theta = 50\sin(53.13^\circ) = 50 \times 0.8 = 40 \text{ N}$$

Solution to Problem 17

Velocity of a Ball

Velocity $v = 12$ m/s at an angle $\theta = 40^\circ$:

(a) $v_x = 12\cos 40^\circ \approx 12 \times 0.766 = 9.19$ m/s.

(b) $v_y = 12\sin 40^\circ \approx 12 \times 0.643 = 7.71$ m/s.

(c) Because the vector splits into two perpendicular dimensions according to sine and cosine, and the sum of their squares equals the square of the magnitude rather than their simple algebraic sum.

1.6 Geometric Applications

Solution to Problem 18

A Ladder Against a Wall

Ladder length $L = 5$ m, distance from base $x = 3$ m.

(a) Vertical height h (via Pythagorean theorem):

$$h = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ m}$$

(b) Angle with the ground θ_{ground}:

$$\cos\theta_{\text{ground}} = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{3}{5} = 0.6 \implies \theta_{\text{ground}} = \cos^{-1}(0.6) \approx 53.13^\circ$$

(c) Angle with the wall θ_{wall} (its complement):

$$\theta_{\text{wall}} = 90^\circ - 53.13^\circ = 36.87^\circ$$

Solution to Problem 19 — A Sloping Ramp

Ramp length $L = 6$ m, vertical rise $h = 1.5$ m.

(a) Angle of inclination above the horizontal θ:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1.5}{6} = 0.25 \implies \theta = \sin^{-1}(0.25) \approx 14.48^\circ$$

(b) Horizontal distance d:

$$d = \sqrt{6^2 - 1.5^2} = \sqrt{36 - 2.25} = \sqrt{33.75} \approx 5.81 \text{ m}$$

(c) The most direct ratio is sine (sin) since the opposite side and hypotenuse are given.

Solution to Problem 20 — Angle of Elevation

Tower height $h = 20$ m, angle of elevation $\theta = 35^\circ$.

(a) Horizontal distance from observer to the base d:

$$\tan 35^\circ = \frac{\text{opposite}}{\text{adjacent}} = \frac{20}{d} \implies d = \frac{20}{\tan 35^\circ} \approx \frac{20}{0.7002} \approx 28.56 \text{ m}$$

(b) Distance from observer to the top of the tower (hypotenuse s):

$$\sin 35^\circ = \frac{20}{s} \implies s = \frac{20}{\sin 35^\circ} \approx \frac{20}{0.5736} \approx 34.87 \text{ m}$$

(c) Triangle sides identification: the tower is the opposite side, the ground distance is the adjacent side, and the line of sight to the top is the hypotenuse.

1.7 Vectors and Trigonometric Synthesis

Solution to Problem 21 — Components and the Fundamental Identity

From the Pythagorean theorem on the vector and its components:

$$A^2 = A_x^2 + A_y^2$$

Substituting the component expressions in terms of angle and magnitude ($A_x = A\cos\theta$ and $A_y = A\sin\theta$):

$$A^2 = (A\cos\theta)^2 + (A\sin\theta)^2 = A^2\cos^2\theta + A^2\sin^2\theta = A^2(\cos^2\theta + \sin^2\theta)$$

Dividing both sides by A^2 (assuming $A \neq 0$):

$$\sin^2\theta + \cos^2\theta = 1$$

as required.

Solution to Problem 22 — Perpendicular Forces

Forces $F_1 = 8$ N and $F_2 = 6$ N are perpendicular.

(a) Magnitude of the resultant R (hypotenuse):

$$R = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \text{ N}$$

(b) Angle with the 8 N force (adjacent):

$$\tan\theta = \frac{6}{8} = 0.75 \implies \theta = \tan^{-1}(0.75) \approx 36.87^\circ$$

(c) Because the two forces represent perpendicular sides of a right triangle, the Pythagorean theorem and trigonometric ratios apply directly to find the magnitude and direction.

Solution to Problem 23 — A Vector with Related Components

Magnitude $A = 10$, with $A_x = 2A_y$.

(a) Relation between sine and cosine: Since $A_x = A\cos\theta$ and $A_y = A\sin\theta$, and $A_x = 2A_y$:

$$A\cos\theta = 2A\sin\theta \implies \cos\theta = 2\sin\theta \implies \cot\theta = 2 \text{ or } \tan\theta = \frac{1}{2}$$

(b) Finding the angle θ:

$$\theta = \tan^{-1}(0.5) \approx 26.57^\circ$$

(c) Finding both components:

$$A_x = 10\cos(26.57^\circ) = 10 \times 0.8944 = 8.94$$

$$A_y = 10\sin(26.57^\circ) = 10 \times 0.4472 = 4.47$$

Solution to Problem 24 — Area and Hypotenuse

Triangle area $A = 24\ \text{cm}^2$ and hypotenuse $c = 10\ \text{cm}$ (meaning $a^2 + b^2 = 100$).

(a) The area of a right triangle is half the product of its legs (a and b):

$$\frac{1}{2}ab = 24 \implies ab = 48$$

Using binomial expansion identities:

$$(a+b)^2 = a^2 + b^2 + 2ab = 100 + 2(48) = 100 + 96 = 196 \implies a + b = \sqrt{196} = 14$$

Similarly:

$$(a-b)^2 = a^2 + b^2 - 2ab = 100 - 96 = 4 \implies a - b = 2$$

Solving this system (sum 14, difference 2): the leg lengths are $a = 8\ \text{cm}$ and $b = 6\ \text{cm}$ (or vice versa).

(b) The two acute angles:

$$\theta_1 = \tan^{-1}\left(\frac{6}{8}\right) = \tan^{-1}(0.75) \approx 36.87^\circ$$

$$\theta_2 = 90^\circ - 36.87^\circ = 53.13^\circ$$

(c) Because the area and hypotenuse provide a sufficient independent algebraic system of equations (sum of squares and product) to uniquely determine side lengths based on the properties of a right-angled triangle.